

PROJECT PROPOSAL

Learning Management System (LMS)

with SCORM-Compliant Course Delivery, Timetable-Based Access, and Online Assessment

1. Project Overview

The proposed Learning Management System (LMS) is a secure, scalable, and web-based platform designed to manage, deliver, and monitor training and educational programs. The system centralizes course administration, learning content, assessments, scheduling, attendance, reporting, and certification. It supports role-based access for Super Administrators, Administrators, Managers, Non-editing Managers, Instructors, and Students, ensuring that every user has access only to functions relevant to their responsibilities.

2. Project Objectives

To design and develop an enterprise-grade LMS that streamlines learning administration, improves learner engagement, automates assessments and reporting, and provides a secure platform for delivering SCORM-compliant online training.

3. Specific Objectives

- Online course management
- SCORM package support
- Quiz and examination module
- Assignment submission
- Timetable-based course availability
- Student progress monitoring
- Attendance monitoring
- Automated grading
- Certificate generation
- Dashboard analytics
- Reports generation

4. Proposed Solution

Develop a centralized LMS featuring role-based access control, SCORM support, timetable-driven course access, online assessments, assignment management, automated grading, progress monitoring, certificate generation, analytics dashboards, reporting, email notifications, and OTP-based authentication.

5. Scope of the System

- User Management
- Authentication
- Course Management
- SCORM Package Management
- Quiz & Assignment Management
- Course Scheduling
- Attendance & Progress Monitoring
- Certificate Management
- Reports & Analytics
- System Administration

6. Risk Management

The project addresses operational and technical risks through role-based permissions, email OTP verification, encrypted passwords, automated backups, audit logs, secure file validation, disaster recovery planning, and infrastructure designed for scalability and reliability.

7. Security of the System

- Role-Based Access Control (RBAC)
- Email OTP authentication
- Password hashing (bcrypt/Argon2)
- HTTPS encryption
- SQL Injection prevention
- CSRF and XSS protection
- Session timeout
- Audit logging
- Regular database backup and recovery

8. Technology Stack

Component	Technology
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript, AJAX
Backend	PHP 8.3 (Laravel 12)
Database	MySQL 8
Web Server	Apache/Nginx

9. System Flow

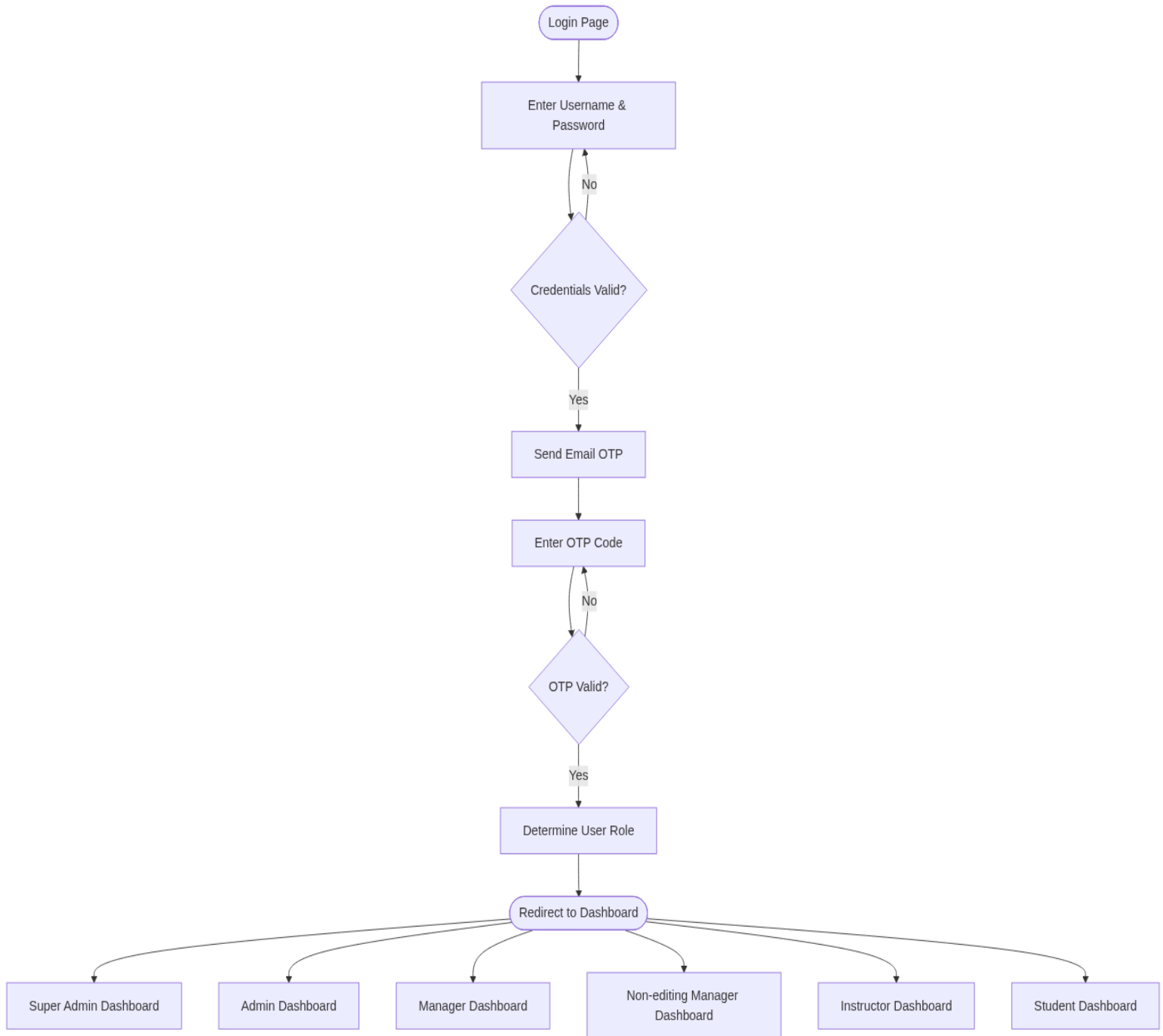


Figure 1: Log in page (General)

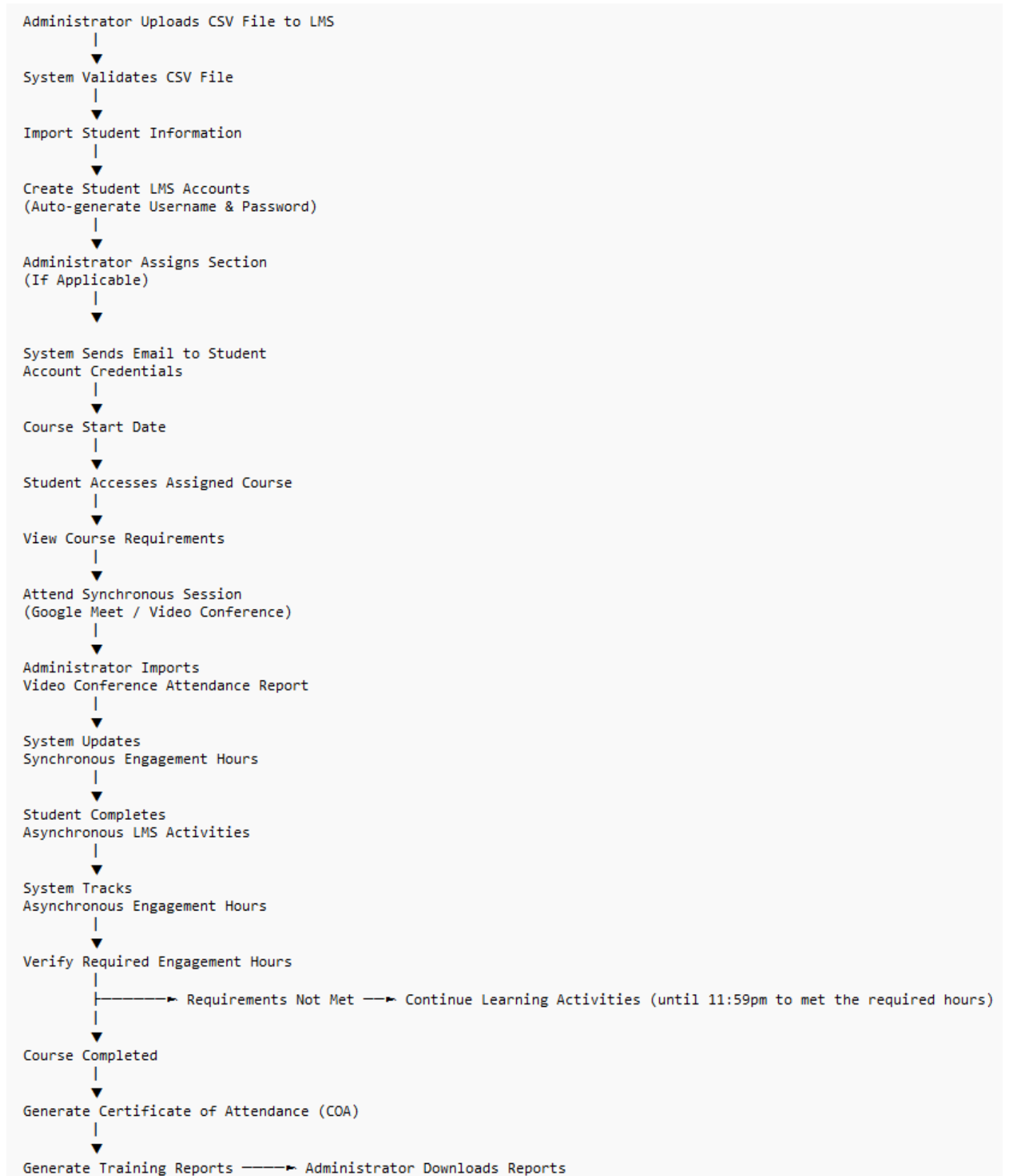


Figure 2: Student's activity inside LMS

Example: BT-PSSR (8-Hour Course)

BT-PSSR (8 Hours)



Required Engagement Time

└─ 4 Hours Synchronous

- Google Meet
- Zoom
- Microsoft Teams

└─ 4 Hours Asynchronous

- SCORM Lessons
- Videos
- Reading Materials
- Quiz

The LMS continuously tracks both engagement types.

Course Progress



Synchronous

4 / 4 Hours ✓

Asynchronous

3.2 / 4 Hours

Total Engagement

7.2 / 8 Hours

Status

Incomplete

Figure 3: Example condition

Completion Rules

The system should only issue a **Certificate of Attendance** if all of the following conditions are satisfied:

- The student has been successfully enrolled and assigned to the course.
- The student has completed the required synchronous training hours (e.g., Google Meet or other approved video conferencing platform).
- The student has completed the required asynchronous learning hours through the LMS (e.g., SCORM course packages, learning modules, videos, and reading materials).
- The student has completed all mandatory learning activities required for the course.
- The student has completed the required quiz or assessment, where applicable.
- The student's total recorded engagement time (combined synchronous and asynchronous hours) meets or exceeds the minimum duration specified for the course.
- All course completion requirements have been validated by the system.

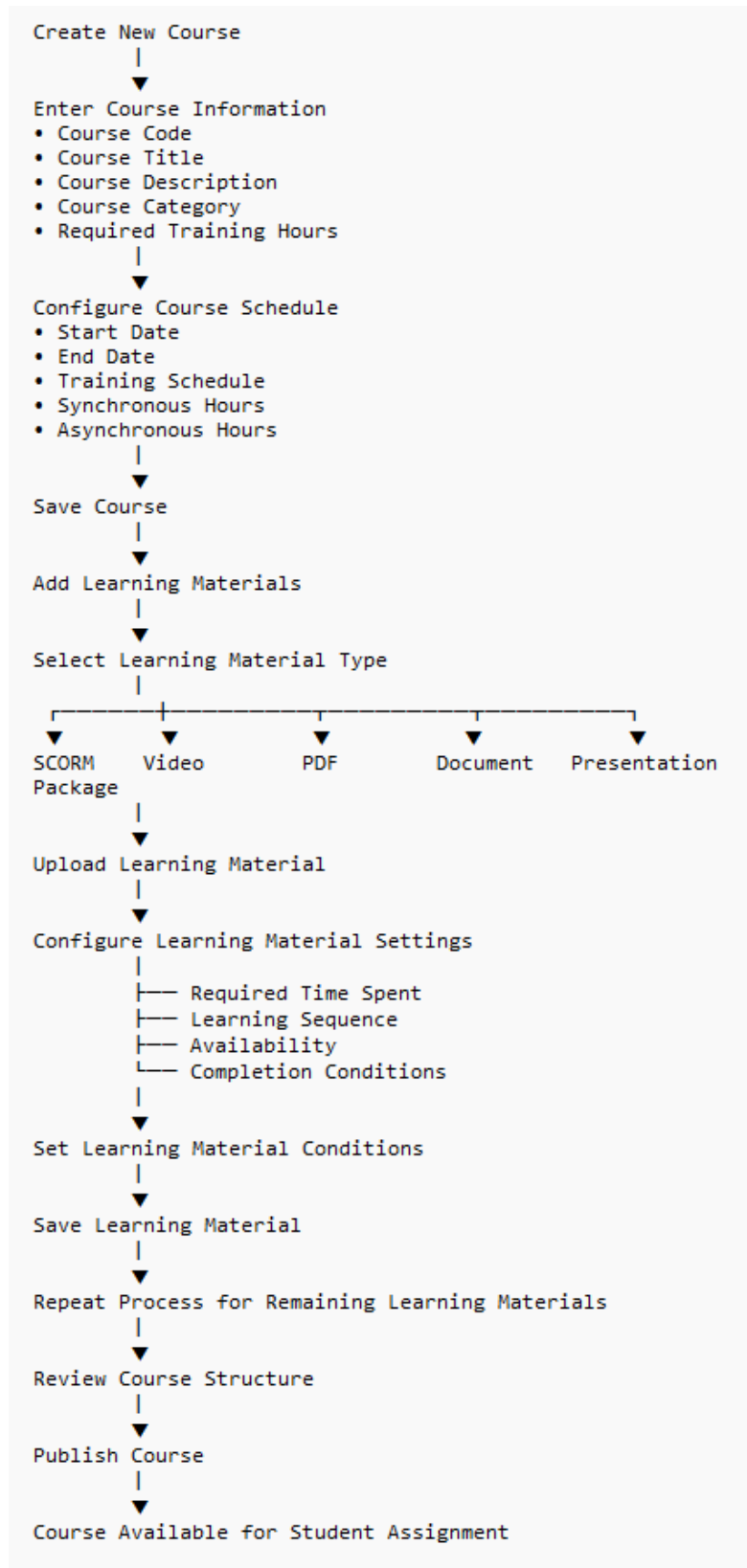


Figure 4: Course Creation and Learning Material Management

Learning Material Completion Logic

For each uploaded learning material, the administrator or instructor may configure one or more completion conditions.

Example

Learning Material 1
SCORM Package

Required Time
30 Minutes

Conditions

- ✓ Scroll to the last page
- ✓ Stay for at least 30 minutes
- ✓ Answer confirmation question
- ✓ Mark as Completed

↓

Unlock Learning Material 2

Learning Material 2
Training Video

Required Time
45 Minutes

Restrictions

- ✓ Cannot Forward Video
- ✓ Cannot Seek Video
- ✓ Cannot Download Video
- ✓ Cannot Copy Video URL

Conditions

- ✓ Watch 100% of the video
- ✓ Stay for at least 45 minutes

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Unlock Learning Material 3

Learning Material 3
PDF Manual

Required Time
20 Minutes

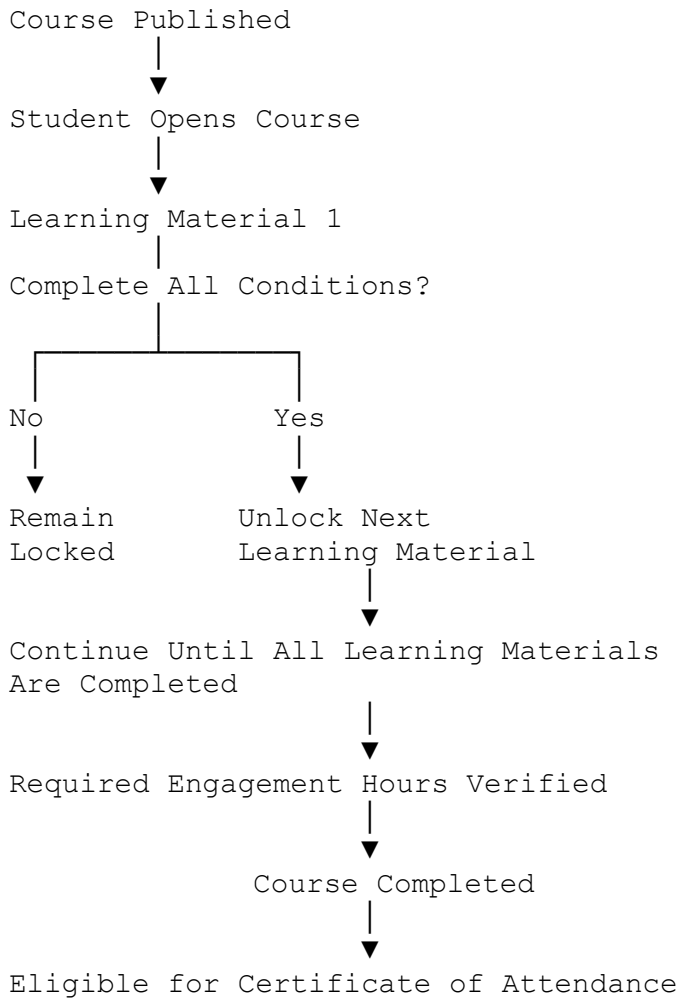
Conditions

- ✓ Scroll to the last page
- ✓ Stay for at least 20 minutes
- ✓ Mark as Read

↓

Unlock Quiz

Course Completion Flow



10. Data Lifecycle Management

Learning Content Storage. All course packages, including SCORM packages, lesson materials (PDF), training videos, exercises, quizzes, and other supporting learning resources, shall be stored on the **web hosting/cloud server**. This centralized repository ensures secure online access, efficient content delivery, and continuous availability of learning materials throughout the training period while supporting simultaneous access by authorized users.

Student Data Retention and Archiving. Student records, including personal information, course enrollments, attendance records, learning progress, assessment results, engagement hours, and Certificate of Attendance (COA) records, shall be maintained in the **web hosting environment** while the course is active. **Thirty (30) days** after course completion, these records shall be archived and transferred to the institution's **on-premises physical server** for long-term retention. The archived records shall be preserved for a period of **ten (10) to fifteen (15) years**, or in accordance with the institution's records retention policy, to support future reference, reporting, and audit requirements.

This version clearly distinguishes between:

- **Learning content** → Permanently stored on the web hosting/cloud server for online delivery.
- **Student records** → Temporarily stored on the web hosting while the course is active, then archived to the institution's physical server for long-term retention.

11. Expected Deployment Requirements

Hardware

- Physical Server (Computer Set)
 - Intel Core i5 (10th Generation or higher)
 - 16 GB RAM
 - 2 × 1 TB SSD (Expandable up to 10 TB)
 - MSI PRO B760M-E Motherboard (or equivalent)
 - 450 W Power Supply (80+ Certified)
 - Computer Chassis with Cooling Fans
- Monitor
- Keyboard
- Mouse
- Uninterruptible Power Supply (UPS)
- External Hard Drive (for backup) *(Recommended)*
- Network Switch *(if multiple devices will be connected)*
- LAN (CAT6) Cables

Web Hosting Services

- Web Hosting
- Domain Name Registration (ex. fareastelearning.ph)
- SSL Certificate (*if not included with the hosting package*)
- Business Email Hosting / Email Server
- Cloud Storage (100 GB or higher)
- Package: <https://www.hostinger.com/pricing>

Learning and Communication Services

- Google Workspace

12. Expected Benefits

- Centralized learning management
- Improved training efficiency
- Automated assessment, engagement tracking, and Certificate of Attendance generation.
- Accurate learner progress monitoring
- Comprehensive reporting and analytics
- Reduced administrative workload
- Secure and scalable platform

13. Future Enhancements

Future releases will integrate with the Enrollment System to automatically synchronize student records and enrollments. Additional enhancements include mobile applications, and video conferencing